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NPS/003/007 – Technical Specification for 66kV and 132kV Disconnectors and Earth Switches

1. Purpose

This document is the technical specification for the procurement of disconnectors and earth switches for use on the CE Electric UK 66kV and 132kV networks.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
NPS/003/007	Rev 1	Technical Specification For 66kV And 132kV Disconnectors And Earth Switches

2. Scope

This specification covers plain break 66kV and 132kV disconnectors for use in CE Electric UK open-compound substations. The specification includes a number of options that may be selected in Appendix-2 depending upon the application of the disconnector or earth switch:

- 1) Manual or motor drive mechanisms
- 2) Bus transfer duty and associated electrical interlocking for use at double-busbar substations
- 3) Reactive switching duty for earth switches required for use on circuits containing long overhead lines.

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3. Technical Requirements

3.1 Technical specification

Disconnectors and earth switches shall comply fully with the Energy Networks Association Technical Specification (ENATS) 41-37 Part 4, as varied in Section 3.2 and Appendices 1 and 2 of this document.

Disconnectors and earth switches must comply with all relevant UK health and safety legislation and will preferably possess Energy Networks Association (ENA) Notice(s) of Conformance.

The following appendices form part of this technical specification.

Appendix 1 details the values, options or amendments selected within ENATS 41-37.

Appendix 2 details project specific options

Appendix 3 contains a schedule of technical specification to be completed by the manufacturer.

Appendix 4 contains a self-certification conformance declaration of compliance with ENATS 41-37.

Appendix 5 contains a self-certification conformance declaration of type test evidence

Appendix 6 contains a declaration sheet of places of manufacture, testing and inspection

3.2 Variations of, or options within ENATS 41-37

This should be read in conjunction with Appendices 1 and 2 of this document.

4.0 Ratings: Rated values of primary and secondary voltage, insulation level, frequency and normal and short circuit currents shall be as specified in Appendices 1 and 2.

4.6 Peak Withstand Current

Where $X/R \Rightarrow 37$ (DC time constant $\Rightarrow 120\text{ms}$) the peak withstand current shall be $2.7 \times$ the short time rated current (i.e. 67.5kA).

4.104 Bus Transfer Duty: Bus transfer duty and associated electrical interlocking will be specified where disconnectors are to be used as busbar disconnectors at double busbar substations. For all other applications busbar transfer duty is not required.

4.105 Induced Current Switching Earth Switches: Earth switches for use on circuits containing long sections of overhead line shall be rated Class B in accordance with Annex C of IEC 62271-102. Earth switches for use on circuits not containing long sections of overhead line shall be rated Class A.

5.11.101 Disconnector and Earth Switch Interlocking: Interlocking arrangements should comply with CEGB Standard 993315 (TPS) 3/9: Interlocking Principles for Transmission Substations and CEGB Standard 993335 (TPS) 3/85: Control Schemes for Open Type 420,300 and 145kV Power Operated Isolators

5.11.101.1 Electrical interlocking: General: Where electrical interlocking inhibits operation of the disconnector it shall inhibit both electrical and mechanical operation unless the mechanical operation is purely for maintenance purposes and locked and labelled as such.

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5.11.102 Key Operated Interlocking: Key interlocks shall be provided as described in (TPS) 3/9. Key interlocks locking off motor drive mechanisms shall also interrupt the motor operating circuit.

4. References

4.1 External Documentation

Reference	Title
ENATS 41-37 Parts 1 and 4	Switchgear for use on 66kV to 132kV Distribution Systems
IEC 62771 – 102	High-Voltage Switchgear and Controlgear – Part 102 Alternating Current Disconnectors and Earthing Switches
CEGB Standard 993315 (TPS) 3/9:	Interlocking Principles For Transmission Substations
CEGB Standard 993335 (TPS) 3/85	Control Schemes for Open Type 420,300 and 145kV Power Operated Isolators

4.2 Internal documentation

Reference	Title
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4.3 Amendments from Previous Version

Section	Subject	Amendments
1.	Purpose	Previous version updated
2.	Scope	Clarification regarding different options within the specification
2.1	Contents	Section added
3..2	Technical Specification: Variations or options within ENATS 41-37	4.0 Ratings: Reference to Appendix 1 of this document
		4.6 Peak withstand current: Increased to 2.7x short time rms value for DC time constants => 120ms
		4.104: Bus transfer duty: Requirement clarified
		4.105: Induced current rating for earth switches: Requirement clarified.
		5.11.101.1: Electrical interlocking: Requirement clarified with regard to CEGB Standard 993315 (TPS) 3/9, Standard 993335 (TPS) 3/85, and also with regard to manual operation of disconnector
		5.11.102: Key Interlocks: Requirement clarified with regard to double busbar substations and also requirement to interrupt motor control circuits.
4.1	External References	Reference to updated

5. Definitions

Term

Definition

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
K.Midgley	Standards Engineer	K.Midgley	18/03/2010

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
Mark Nicholson	Head of System Strategy	Mark Nicholson	29/03/2010

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
D.Blackledge	Plant Standards Manager	D.Blackledge	18/03/2010

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S.Johnson	DBD Administrator	S.Johnson	14/04/2010

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C.Holdsworth	Standards and Assessment Manager	C.Holdsworth	17/03/2010

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C.Morgan	Head of Procurement	C.Morgan	31/03/2010

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Appendix 1: Specified electrical ratings and variations within ENATS 41-37 66kV Disconnecter and Earth Switch

	ENATS 41-37 Reference	72.5kV disconnector requirement
Particulars of system		
Nominal System Voltage		66kV
Frequency		50Hz
Number of phases		3
Disconnecter/earthswitch characteristics		
Number of poles		3
Class: Indoor, outdoor		Outdoor
Rated voltage	Part 1, 4.1	72.5 kV
Rated insulation level: Lightning Impulse Power Frequency	Part 1, 4.2 (Table 1)	325/375kV 140/160kV
Rated Frequency	Part 1, 4.3	50Hz
Rated normal current	Part 1, 4.4	1250A: Feeder cct 2000A: TX, BS and BC cct
Rated short-time withstand current	Part 1, 4.5	25kA
Rated Peak Withstand Current	Part 1, 4.6	62.5kA X/R<37 67.5kA X/R=>37
Rated duration of short circuit	Part 1, 4.7	3 sec
Rated supply voltage of closing and opening devices and control circuits – see table 1.2	Part 1, 4.8	Nominal 110V Max 137.7V Min 87.5V
Rated supply frequency of closing and opening and of auxiliary circuits.	Part 1, 4.8	DC
Classification of mechanical operations Class M0-1000 or M1-2000	Part 2, 4.106	M1-2000
Rated mechanical terminal load IEC 62271- 102 sub clause 4.103	(IEC 62271-102 sub clause 4.103)	Supplier to state
Rated bus transfer switching IEC 62271 102 sub clause 4.104	Part 4, 4.104	80% normal rated current (1600A max) @ 100V bus transfer voltage
Rated induced current switching of Earthing Switches IEC 62271-102 4.105	Part 4, 4.105	Class A: Short OHL lengths. Class B: Long OHL lengths
Rated values of electrical endurance IEC 62271 – 102 subclause 4.107	(IEC 62271 –102 sub clause 4.107)	Class E0
Earthing of disconnectors and earth switches IEC 62271 clause 5.3		N/A
Mechanism type: earth switch		Supplier to state
Mechanism type: disconnector	Part 4, Clause 5	Dependent manual/dependent power
No. of auxiliary contacts	Part 4, 5.4.101	10 NO, 10 NC
Key Interlock type	Part 4, 5.11.102 as expanded by this document.	Supplier to state

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Key Interlocks	Part 4, 5.11.102 as expanded by CEGB Standard 993315 (TPS) 3/9	On motorised mechanisms key interlocks shall also interrupt motor control circuits.
Electrical Interlock	CEGB Standard 993315 (TPS) 3/9 Standard 993335 (TPS) 3/85	Should prevent motor and manual operation.
Padlocking facilities	Part 1, 5.11.102	Supplier to state locking arrangements
Mechanism Degree of Protection	Part 1, 5.13	
Surface preparation		Supplier to state
Mechanism box heater voltage		110V AC

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132kV Disconnecter and Earth Switch

	ENATS 41-37 Reference	132kV disconnector
Particulars of system		
Nominal System Voltage		132kV
Frequency		50Hz
Number of phases		3
Disconnecter/earthswitch characteristics		
Number of poles		3
Class: Indoor, outdoor		Outdoor
Rated voltage	Part 1, 4.1	145 kV
Rated insulation level: Lightning Impulse Power Frequency	Part 1, 4.2 (Table 1)	550/630kV 230/265kV
Rated Frequency	Part 1, 4.3	50Hz
Rated normal current	Part 1, 4.5	1250/2000A
Rated short-time withstand current	Part 1, 4.5	25kA
Rated Peak Withstand Current	Part 1, 4.6	62.5kA X/R<37 67.5kA X/R=>37
Rated duration of short circuit	Part 1, 4.7	3 sec
Rated supply voltage of closing and opening devices and control circuits	Part 1, 4.8	Nominal 110V Max 137.7V Min 87.5V
Rated supply frequency of closing and opening and of auxiliary circuits.	Part 1, 4.8	DC
Classification of mechanical operations Class M0-1000 or M1-2000	Part 2, 4.106	M1-2000
Rated mechanical terminal load	(IEC 62271-102 sub clause 4.103)	Supplier to state
Rated bus transfer switching IEC 62271 102 sub clause 4.104	Part 4, 4.104	80% normal rated current (1600A max) @ 100V bus transfer voltage
Rated induced current switching of Earthing Switches IEC 62271-102 4.105	Part 4, 4.105	Class A: Short OHL lengths. Class B: Long OHL lengths
Rated values of electrical endurance IEC 62271 – 102 subclause 4.107	(IEC 62271 –102 sub clause 4.107)	Class E0
Earthing of disconnectors and earth switches IEC 62271 clause 5.3		N/A
Mechanism type: disconnector	Part 4, Clause 5	Dependent manual/dependent power
Mechanism type: earth switch		Dependent manual
No of auxiliary contacts	Part 4, 5.4.101	10 NO, 10NC
Key Interlock type	Part 4, 5.11.102 as expanded by this document.	Supplier to state
Key Interlock	Part 4, 5.11.102 as expanded by CEGB Standard 993315 (TPS) 3/9	On motorised mechanisms key interlocks shall also interrupt motor control

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Electrical Interlock	CEGB Standard 993315 (TPS) 3/9 Standard 993335 (TPS) 3/85	Should prevent motor and hand operation.
Padlocking facilities	Part 1,5.11.102	Supplier to state locking arrangements
Mechanism Degree of Protection	Part 1, 5.13	
Surface preparation		Supplier to state
Mechanism box heater voltage		110V AC

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Appendix 2 – Project Specific Technical Requirements

Summary of project specific technical requirements within the context of Section 3.2 and Appendix 1

Information	ENATS 41-37 Reference	Enquirer to indicate requirement as appropriate
Rated Voltage. See Appendix 1.	Part 1, 4.1	72.5kV / 145kV
Normal rated current. See Appendix 1.	Part 1, 4.5	1250A / 2000A
Bus-transfer and associated electrical interlocking (TPS 3/9). Refer section 3.2 and Appendix 1.	Part 4, 4.104	Yes / No
Rated induced current switching of earth switch see Section 3.2 and Appendix 1.	Part 4, 4.105	Class A / Class B
X/R (DC time constant) of network	Part 1, 4.6	$X/R < 37$ / $X/R \Rightarrow 37$
Disconnecter operating mechanism: Motor or manual		Motor / Manual
Earth switch operating mechanism: Motor or manual		Motor / Manual
Key interlocks		State requirement
Electrical Interlocking		State any special requirements requirement:

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Appendix 3: Technical schedule to be completed by the manufacturer

	ENATS 41-37 Ref	
Disconnecter		
Manufacturer		
Manufacturer's Designation		
Type (e.g RCP, ROP, Pantograph etc)		
ENA Notice of Conformity	No.	
Number of poles		
Class: Indoor, outdoor		
Rated voltage	Part 1, 4.1	
Rated insulation level (lightning impulse withstand voltage) see table 1a IEC 60694	Part 1, 4.2	
Rated Frequency	Part 1, 4.3	
Rated normal current	Part 1, 4.5	
Rated short-time withstand current	Part 1, 4.6	
Rated duration of short circuit	Part 1, 4.7	
Classification of mechanical operations Class M0-1000 or M1-2000	Part 2, 4.106	
Rated mechanical terminal load	IEC 62271-102 clause 4.103	
Rated bus transfer switching	Part 4, 4.104	
Rated values of electrical endurance	IEC 62271-102 clause 4.107	
Disconnecter Mechanism		
Manufacturer's Designation		
Mechanism type: Dependent manual/dependent power		
Rated supply voltage of closing and opening devices and control circuits		
Rated supply voltage of control circuits		
Motor rating (W)		
Operating time of opening and closing (s)		
Motor protection		
Manual operation: method of interlock with motor mechanism		
Manual operation: method of interlock with substation electrical interlock circuits.		
No of auxiliary contacts	Part 4, 5.4.101	
Key Interlock type	Part 4, 5.11.102 as expanded in this document	
Key Interlock: Method of interlock with motor control circuits.		
Electrical Interlocking	TPS 3/85	
Padlocking facilities	Part 1, 5.11.102	
Mechanism Degree of Protection	Part 1, 5.13	
Surface preparation		
Mechanism box heater voltage		

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	ENATS 41-37 Ref	
Disconnecter Weights		
Mass of disconnector (minus mechanism) (kg)		
Mass of Mechanism (kg)		
Disconnecter Drawings		
Disconnecter General Arrangement		
Operating mechanism layout		
Control Schematic		
Earth Switch		
Manufacturer		
Manufacturer's Designation		
Type (e.g chopper, pantograph etc)		
Number of poles		
Class: Indoor, outdoor		
Rated voltage	Part 1, 4.1	
Rated insulation level (lightning impulse withstand voltage) see table 1a IEC 60694	Part 1, 4.2	
Rated Frequency	Part 1, 4.3	
Rated short-time withstand current	Part 1, 4.4	
Rated induced current switching of Earthing	Part 4, 4.105	
Rated duration of short circuit	Part 1, 4.5	
Classification of mechanical operations Class M0-1000 or M1-2000	Part 2, 4.106	
Earthswitch Mechanism		
Manufacturer's Designation		
Mechanism type: Dependent manual/dependent power		
No of auxiliary contacts	Part 4, 5.4.101	
Key Interlock type	Part 4, 5.11.102 as expanded in this document	
Padlocking facilities	Part 1, 5.11.102	
Mechanism Degree of Protection	Part 1, 5.13	
Surface preparation		
Mechanism box heater voltage		
Earth Switch Weights		
Mass of earth (minus mechanism) (kg)		
Mass of Mechanism		
Earth Switch Drawings		
Earth Switch General Arrangement		
Operating mechanism layout		

Appendix 4 - Self-certification declaration of conformance to ENATS 41 - 37

Self Certification Conformance Declaration

Disconnectors and Earthing Switches

CLAUSE BY CLAUSE CONFORMANCE WITH ENA TS 41-37 – Part 4 ADDITIONAL CLAUSES FOR High Voltage Disconnectors and Earthing switches Switchgear covered by ENA TS 41-37 shall comply with the latest issues of the relevant International and British Standards. ENA TS 41-37 is intended to amplify and/or clarify the requirements of those Standards.

This check sheet identifies the clauses in ENA TS 41-37 - Part 2 and the clauses of the aforementioned Standards relevant to clauses High voltage Disconnectors and Earthing switches.

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable or appropriate to the product

Cs1 = the product is fully conforms with the requirements of this clause

Cs2 = the product partially conforms with the requirements of this clause

Cs3 = the product does not conform with the requirements of this clause

Cs4 = the product does not currently conform with the requirements of this clause, but the
Manufacturer proposes to modify and test the product in order to comply.

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'IEC' or 'ENA TS' as appropriate

Manufacturer:

Ratings:

Product Reference:

Signature:

Date:

IEC 62271-203, & IEC 62271-102				ENA TS 41-37			
Clause / Subclause		Requirement	Conformance code	ENA TS 41-37 - Part 4 Clause / Subclause	Requirement	Conformance code	Remarks
Part I of 41-37	IEC62271-102						
2	2	Normal and special service conditions		2	Normal and special service conditions		
3	3	Definitions		3	Definitions		
4	4	Ratings		4	Ratings		
	4.104.			4.104	Bus transfer		
	4.105			4.105	Induced current switching Line earthing switches		
	4.106			4.106	Mechanical endurance		
				4.201	Earthing Devices having a rating short circuit making current		
1.5	5	Design and construction		5	Design and construction		
5.4	5.4	Auxiliary and control equipment		5.4	Auxiliary and control equipment		
				5.4.101	Setting adjustment of Auxiliary switches		
5.11	5.11	Interlocking devices		5.11	Interlocking and padlocking facilities		
				5.11.101	Disconnecter and earthing switch interlocking		
				5.11.102	Key operated interlocking		
				5.11.102.1	Interlock key release position		
				5.201	Disconnecter three- pole operation		
				5.202	Operations counter		

IEC 62271-203, & IEC 62271-102				ENA TS 41-37			
Clause / Subclause		Requirement	Conformance code	ENA TS 41-37 - Part 4 Clause / Subclause	Requirement	Conformance code	Remarks
Part I of 41-37	IEC62271-102						
				5.203	Manually operation of disconnectors with bus-transfer duty		
				5.204	Installation tolerances		
6	6	Type tests		6	Type tests		
	6.102.3			6.102.3	Mechanical endurance tests		
7	7	Routine tests		7	Routine tests		
8	8	Guide to selection of disconnectors and earthing switches		8	Guide to selection of disconnectors and earthing switches		
9	9	Information to be given with enquires, tenders and orders		9	Information to be given with enquires, tenders and orders		
10	10	Rules for transport, storage, installation, operation and maintenance		10	Rules for transport, storage, installation, operation, maintenance and disposal		
				10.101	Assembly instructions		
11	11	Safety		11	Safety		
	12	Environmental Aspects		12	Environmental aspects		

Appendix 5 – EATS 41-37 Type test self certification conformance declaration

High Voltage Disconnectors and Earthing Switches

Manufacturer:

Ratings.....

Product reference:

Name:

Signature:

Date:

Instructions for completion:

- Complete a separate table for each variant and rating
 - ENA/SAP to complete columns 1 to 4
 - Manufacturer to complete columns 5 to 10
 - When test report also covers another rating insert 'See ???A unit' in the Remarks column
- Tests not requested may be shown as 'Additional tests' at the bottom of the table

*See bottom of table for conformance declaration codes

** I = Independent; M= Manufacturer; ENA= Energy Networks Association

	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
	Test Requirement	Specification and Standards	Rated value	Test req'd Y or N	Conformance	Test value	Date of test	Test station Report / Cert No	Witness I, M or ENA**	Remarks
1	Dielectric	Subclause 6.2. IEC 62271-102.								
2	Radio interference test	Subclause 6.3 IEC 60694								
3	Measurement of the resistance of main circuit -	Subclause 6.4 IEC 60694								
4	Temperature Rise	Subclause 6.5, IEC 60694								
5	Short-time withstand current and peak withstand current tests -	Subclause 6.6 IEC 62271-102 and IEC 60694 Subclause 6.6.1								
6	Verification of protection	Subclause 6.7 IEC 60694								

CAUTION! - This document may be out of date if printed

	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
	Test Requirement	Specification and Standards	Rated value	Test req'd Y or N	Conformance	Test value	Date of test	Test station Report / Cert No	Witness I, M or ENA**	Remarks
7	Tightness test	Subclause 6.8 IEC 60694								
8	EMC tests	Subclause 6.9 IEC 60694								
9	Tests to prove the short circuit making performance of earthing switches.	Subclause 6.101 IEC 62271-102								
10	Operating and mechanical endurance test	Subclause 6.102 IEC 62271-102								
11	Operation under severe ice conditions	Subclause 6.103 IEC 62271-102								
12	Operation at the temperature limits	Subclause 6.104 IEC 62271-102								
13	Tests to prove the proper function of the position –indicating device	Subclause 6.105 IEC 62271-102								
14	Bus-transfer current switching test	Subclause 6.106 IEC 62271-102								
15	Induced current switching tests	Subclause 6.107 IEC 62271-102.								
16	Bus-charging switching tests	Subclause 6.108 IEC 62271-102								
17	Additional tests on auxiliary and control circuits	Subclause 6.10 IEC 60694								
18	Routine tests	Clause 7 IEC 62271-102								
18	Finish	Performance to ENA TS 98-1								
19	Process Control	ISO 9000 ERG79 Parts 1 & 2a								

	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
	Test Requirement	Specification and Standards	Rated value	Test req'd Y or N	Conformance	Test value	Date of test	Test station Report / Cert No	Witness I, M or ENA**	Remarks
Note All type tests carried out using the number of test samples specified in I 6.1.1 of IEC 60694. Disconnectors and earthing switches with a rated voltage of 72.5kV and above tests marked by * an additional test sample is allowed for the marked test.										

*** Conformance declaration codes**

N/A = Clause is not applicable/appropriate to the product

Cs1 = the test conforms fully with the requirements of this clause

Cs2 = the test conforms partially with the requirements of this clause

Cs3 = the test does not conform to the requirements of this clause

Cs4 = Test not performed, but alternative evidence/ technical case offered

Ct1 = Independent witnessed tests

Ct2 = Not fully independent witnessed tests

Ct3 = Self verified tests

Ct4 = Alternative tests / evidence offered

Ct5 = Manufacturer has underwritten that the product meets the functional and performance requirements without further testing.

Ct6 = Not tested

Appendix 6 – Manufacturer, places of manufacture, testing and inspection

Item	Manufacturer's Drawing Number and/or Type Designation	Manufacturer	Place of Manufacture	Place of Testing and Inspection
Disconnecter primary components				
HV Insulators				
Operating mechanism				
Support Steelwork				
Any deviation from this Schedule shall be notified in writing as soon as possible for the Purchaser's approval.				